-A Simulation-Based Analysis of a Nozzle Expansion Efficiency Proxy on Small-Scale Rocket Performance

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Abstract

The behavior of nozzle expansion plays a crucial role in determining the thrust efficiency of a rocket and its overall performance, especially for small-scale vehicles operating under varying atmospheric conditions. This study investigates the effects that nozzle expansion behavior has on flight performance using OpenRocket simulations. As modificating nozzle expansion is limited within the software, a thrust efficiency multiplier was used as a simplified representation of changes in expansion behavior while maintaining constant vehicle geometry, mass properties, motor selection, and atmospheric conditions. Simulations were conducted across several thrust multiplier values with repeated trials to account for stochastic wind conditions. Key performance metrics, including apogee, maximum velocity, and time to apogee, were recorded and analyzed after each trial. Results showed a consistent increase in apogee with increasing effective expansion behavior, while maximum velocity and time to apogee exhibited relatively small variation. These findings suggest that small-scale rocket performance is sensitive to changes in the thrust-efficiency proxy used to represent nozzle expansion behavior. This study is exploratory in nature and does not include experimental validation; results are intended only to illustrate performance trends.

1. Introduction

Rocket nozzle expansion behavior plays a very important role in thrust efficiency and overall rocket performance, as it determines how effectively exhaust gases expand to match ambient pressure. Rocket propulsion theory shows that when nozzle expansion does not match operating conditions, performance losses occur, especially through reduced thrust efficiency and specific impulse. If exhaust gases are under-expanded or overexpanded, some of the available energy is not converted into useful thrust. These effects can become more apparent as operating conditions change during flight, for example with increasing altitude.

In small-scale rocket systems, nozzle geometry is typically fixed by commercial motor design, which limits opportunities for direct experimental variation. As a result, simulation-based tools are better used to explore how changes in nozzle expansion behavior affects flight performance. Software such as OpenRocket is already widely used in educational and amateur rocketry settings to model rocket trajectories and propulsion effects under controlled conditions.

This study investigates how changes in a thrust-efficiency proxy used to represent nozzle expansion behavior influence small-scale rocket flight performance. As direct modification of nozzle geometry is limited within the software, a thrust efficiency multiplier is used as a simplified proxy to represent changes in expansion behavior while maintaining constant vehicle geometry, mass properties, motor selection, and atmospheric conditions. The objective of this study is to examine how variations in the expansion-behavior proxy affect key performance metrics, including apogee, maximum velocity, and time to apogee, under simulated conditions.

2. Methods

2.1 Simulation Environment

All flight simulations were done using OpenRocket, which is an open-source rocket flight simulation software. OpenRocket was selected due to its ability to model small-scale rocket flight dynamics, including thrust curves, aerodynamic drag, mass properties, and atmospheric conditions, while allowing controlled parametric studies.

2.2 Baseline Rocket Configuration

A baseline rocket model was made in OpenRocket and held as a constant throughout the study. The vehicle featured a cylindrical body tube (approximately 24 mm diameter, 36 cm length), a pointed nose cone, and a three-fin trapezoidal fin configuration. Total mass, center of gravity, center of pressure, and overall geometry were fixed for all simulations to ensure that observed performance differences were caused solely by propulsion-related parameters.

A commercially available C6-5 motor was used for all of the simulations. Launch angle was set to 90° (vertical), and wind conditions were constrained to low values (0–2 m/s). Atmospheric conditions, launch rod configuration, and recovery settings were also constants across all trials.

2.3 Nozzle Expansion Proxy

OpenRocket does not permit direct modification of nozzle geometry or physical expansion geometry within motor definitions. To study how nozzle expansion affects rocket performance, a thrust efficiency multiplier (ε) was used as a simplified proxy. This multiplier scales the effective thrust output of the motor while preserving the original thrust curve shape.

Three thrust efficiency values were examined:

$\varepsilon = 0.95$

$\varepsilon = 1.00$ (baseline)

$\varepsilon = 1.05$

All simulations maintained identical vehicle geometry, mass properties, motor selection, and atmospheric conditions. Only the thrust efficiency multiplier was adjusted, allowing a controlled assessment of performance sensitivity to effective nozzle expansion behavior.

2.4 Simulation Procedure

For each ε value, five independent simulation trials were executed to account for variation within the simulation environment. Output data was exported directly from OpenRocket without modification. The following performance metrics were recorded for each trial:

Apogee (m)

Maximum velocity (m/s)

Time to apogee (s)

Average values for each metric were computed using the five trials per ε condition. One baseline trial produced an anomalously high apogee value and was excluded from averaged results while remaining documented in the raw dataset (see Section 6 and Appendix C).

2.5 Data Processing and Visualization

Simulation outputs were organized and analyzed using Google Sheets. Average performance metrics were compiled into comparison tables, and graphical visualizations were generated to illustrate the trends in apogee, maximum velocity, and time to apogee as functions of ε. Final figures were then exported for inclusion in the Results section.

3. Results

3.1 Apogee Performance

Figure 1 presents the average apogee achieved as a function of the thrust efficiency multiplier (ε). A modest but noticeable increase in apogee was observed with increasing ε across the tested range. The baseline configuration (ε = 1.00) achieved an average apogee of approximately 49.6 m, while the ε = 0.95 and ε = 1.05 configurations resulted in average apogees of approximately 49.4 m and 50.0 m, respectively.

The observed trend suggests that increased thrust efficiency, used here as a proxy for nozzle expansion behavior, leads to higher peak altitude under identical flight conditions. However, the magnitude of the increase is small, indicating limited sensitivity of apogee to ε within the tested range.

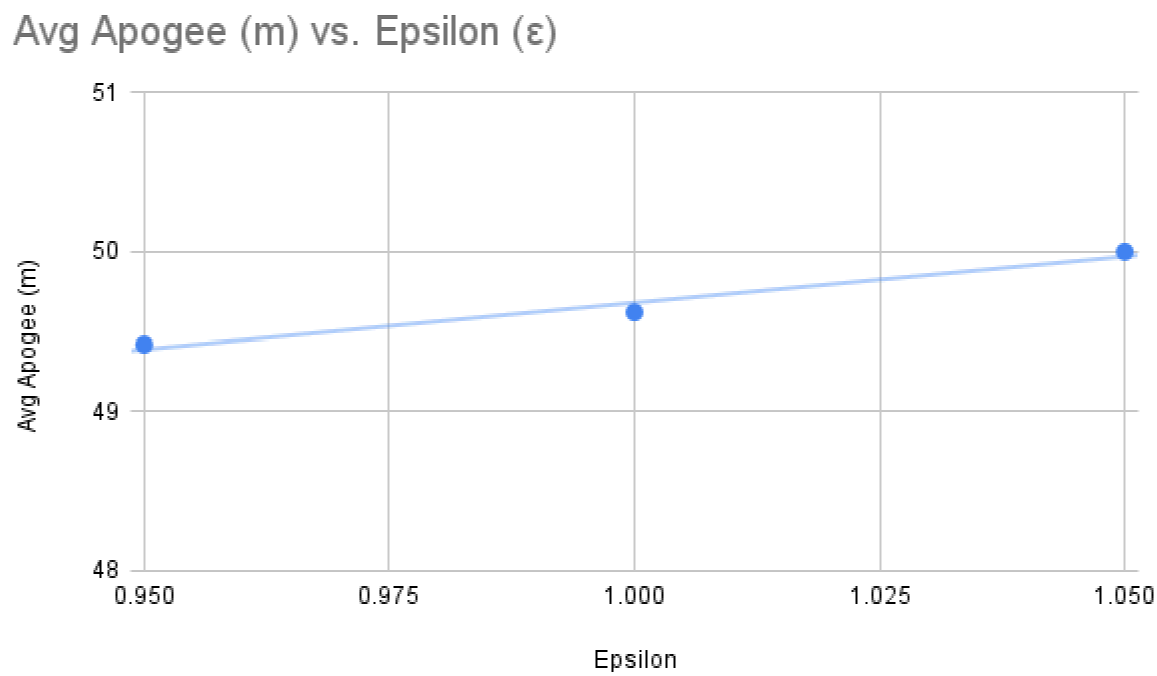

Figure 1. Average apogee as a function of thrust efficiency multiplier (ε) obtained from OpenRocket simulations. Apogee showed a modest increase as the thrust efficiency multiplier (ε) increased across the tested range.

3.2 Maximum Velocity

Figure 2 shows the relationship between average maximum velocity and the thrust efficiency multiplier (ε). Across all configurations, the average maximum velocity remained relatively constant at around 83–84 m/s. No strong trend with ε was observed.

This result indicates that while small changes in thrust efficiency may influence overall altitude performance, peak velocity is largely governed by motor characteristics and vehicle mass rather than nozzle expansion effects at this scale. The minimal variation in maximum velocity supports the conclusion that ε primarily affects integrated flight performance rather than instantaneous peak acceleration.

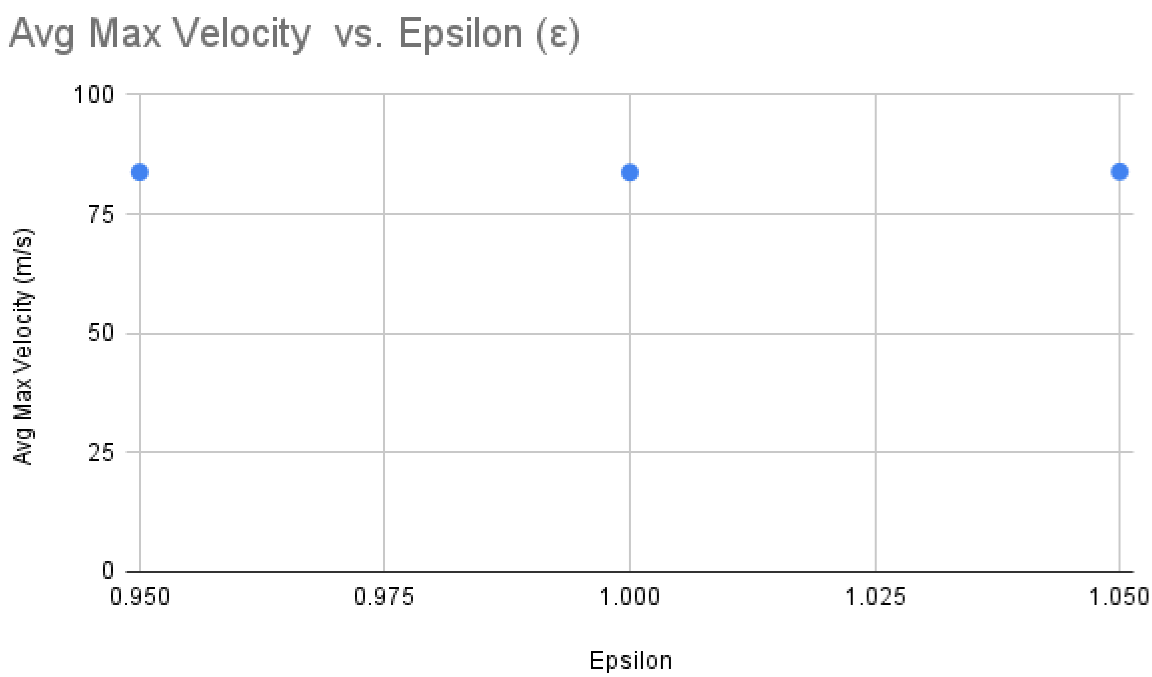

Figure 2. Average maximum velocity as a function of thrust efficiency multiplier (ε). Maximum velocity shows minimal sensitivity to ε.

3.3 Time to Apogee

The average time to apogee for all configurations ranged from approximately 3.05 s to 3.22 s. The ε = 1.05 configuration exhibited a slightly longer time to apogee compared to the baseline case, consistent with its modestly increased apogee altitude. The ε = 0.95 and ε = 1.00 configurations showed nearly identical times to apogee.

Overall, time to apogee demonstrated minimal sensitivity to variations in ε within the tested range. The results suggest that changes in thrust efficiency affect total ascent duration primarily through small differences in achieved altitude rather than substantial changes in ascent dynamics.

3.4 Summary of Results

In all simulations, increasing the thrust efficiency multiplier resulted in:

A modest increase in average apogee

Minimal change in maximum velocity

Slight variation in time to apogee consistent with altitude differences

The findings indicated that nozzle expansion effects, which are as modeled through a thrust efficiency proxy, exert a measurable but limited influence on small-scale rocket flight performance under the simulated conditions.

4. Discussion

The results of this simulation-based study suggest that variations in the thrust efficiency multiplier, used here as a proxy for nozzle expansion behavior, have a measurable but limited effect on the flight performance of a small-scale rocket under otherwise identical conditions.

A modest increase in average apogee was observed as the thrust efficiency multiplier increased. This indicates that improved thrust efficiency is associated with slightly greater integrated impulse during ascent. This general trend is qualitatively consistent with classical rocket propulsion theory, in which more effective nozzle expansion increases the conversion of chamber pressure into usable thrust. However, the magnitude of the observed apogee increase was small across the tested range of efficiency values (0.95–1.05), indicating that changes in effective nozzle expansion may produce diminishing returns for altitude performance at this scale.

In contrast, maximum velocity showed minimal sensitivity to changes in the thrust efficiency multiplier. Peak velocity values remained similar across all configurations, suggesting that maximum speed during ascent is influenced more strongly by motor characteristics, vehicle mass, and aerodynamic drag than by small variations in effective nozzle expansion. As maximum velocity occurs early in flight when thrust is highest and drag forces increase rapidly, small efficiency changes appear insufficient to significantly alter this behavior.

Time to apogee also exhibited only slight variation across configurations and closely followed changes in achieved altitude. The highest efficiency configuration reached apogee marginally later, which is consistent with its slightly greater altitude. These differences were small and did not indicate a separate or independent trend.

Overall, the findings suggest that changes in thrust efficiency influence integrated flight performance metrics, such as total altitude, more noticeably than instantaneous metrics like maximum velocity. While nozzle expansion behavior plays a role in small-scale rocket performance, its influence is constrained by other more dominant factors such as aerodynamic drag, vehicle mass, and motor selection. Under the simulated conditions used in this study, improvements in effective nozzle expansion alone are unlikely to result in large performance gains.

5. Limitations

This study was conducted entirely using numerical simulations and therefore has several limitations common to simulation-based analysis. While OpenRocket is a widely used tool for amateur and educational rocketry analysis, simulated results may not fully represent real-world flight behavior due to modeling assumptions and idealized conditions.

First, nozzle expansion behavior was not modeled directly through changes to nozzle geometry. Instead, a thrust efficiency multiplier (ε) was used as a simplified proxy to represent changes in effective expansion efficiency. Although this approach allows controlled variation of propulsion efficiency while keeping vehicle geometry and mass properties constant, it simplifies the complex flow and thermodynamics behavior inside a rocket nozzle. As a result, the modeled ε values may not correspond directly to physically realizable nozzle geometries.

Second, the simulations assumed ideal and repeatable environmental conditions, including constant atmospheric conditions and the absence of wind, launch misalignment, or manufacturing tolerances. In real flight scenarios, these factors can introduce additional variability that may reduce or mask small performance differences related to nozzle expansion effects.

Third, no experimental validation was performed. The absence of physical flight-testing limits how confidently the observed trends can be confirmed and how accurately the thrust efficiency proxy represents real nozzle performance. Future work incorporating static motor testing or flight experiments would be necessary to validate the simulation results and extend the applicability of the conclusions.

Finally, the analysis was limited to a narrow range of ε values and a single vehicle and motor configuration. As a result, the findings may not directly apply to larger rockets, higher Mach number limits, or propulsion systems operating under different pressure conditions.

Despite these limitations, the study provides a useful preliminary investigation into the relative influence of nozzle expansion efficiency on small-scale rocket performance and demonstrates the educational and exploratory value of simulation-based methods.

6. Outlier Handling and Data Integrity

All simulation trials conducted in this study were kept in the raw dataset to support transparency and reproducibility. During the baseline simulation runs, one trial produced an unusually high apogee value of 193 m, which was higher than the other baseline trials under identical simulation conditions.

This data point was identified as an outlier because it deviated from the main cluster of baseline results and was not representative of typical system behavior. To avoid negatively impacting average performance metrics and comparative trends, this outlier was excluded from mean calculations and graphs, but it was not deleted or modified and remains fully documented in the Appendix, with raw data tables available upon request.

No simulations were rerun, altered, or selectively removed to influence results. All subsequent analyses were performed using the remaining consistent baseline trials, which exhibited stable and repeatable results. The same data handling methodology was applied consistently across all thrust efficiency multiplier (ε) conditions.

This approach follows common research practice by balancing transparency with integrity and ensures that reported trends reflect typical system behavior rather than anomalous results.

7. Conclusion

This study explored how effective nozzle expansion behavior influences small-scale rocket performance using simulation-based analysis. A thrust efficiency multiplier was used as a proxy for effective nozzle expansion behavior, allowing controlled variations while keeping vehicle geometry, propulsion hardware, and atmospheric conditions constant. The results show that increases in effective nozzle expansion behavior are associated with modest increases in apogee, while maximum velocity and time to apogee changed very little across the tested range.

Although the analysis is limited to simulations and proxy modeling, the observed trends are consistent with classical rocket propulsion principles related to thrust efficiency and exhaust expansion. This study highlights the usefulness of simulation based tools such as OpenRocket for exploring propulsion-related concepts in small-scale rocket systems. Future work may include experimental testing or higher-fidelity modeling to examine nozzle expansion effects under different flight conditions.

8. References

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9. Appendix

The Appendix provides supplementary materials to support transparency, reproducibility, and methodological clarity. All materials referenced here were generated during the study and were used directly in the analysis presented in this paper.

Appendix A: OpenRocket Configuration Screenshots

Screenshots of the baseline and modified OpenRocket configurations were used to document vehicle geometry, motor selection, launch conditions, and simulation parameters used throughout the study.

Appendix B: Raw Simulation Data Tables

Raw simulation output data were generated for all baseline and thrust efficiency multiplier (ε) trials and were used during analysis to compute averaged performance metrics. These data include individual trial values for apogee, maximum velocity, and time to apogee. Raw data tables are not included in this PDF but are available upon request.

Appendix C: Outlier Handling Documentation

This appendix documents the identification and handling of an anomalous 193 m apogee value observed in one baseline simulation trial. The criteria used to identify this outlier and the rationale for excluding it from averaged results are described. The raw baseline dataset was examined during analysis but is not included in this PDF and is available upon request.

Appendix D: Project Folder Structure Summary

During this study, simulation outputs, figures, intermediate analysis files and documentation were organized locally using a structured folder system to support systematic analysis and version control. The directory structure is not included in this PDF release and it was used to separate simulation outputs, processed results, figures, and manuscript materials during the research process. Supporting files are available upon request.

Appendix E: OpenRocket Displayed Angle Clarification

During simulations that included wind and turbulence modeling, OpenRocket displayed an apparent flight angle of approximately 60° in some modified trials. This displayed value does not represent the physical launch rod angle, which was maintained at 90° (vertical) for all simulations. The displayed angle reflects the effective airflow-relative orientation of the rocket resulting from wind alignment and weathercocking behavior rather than a change in launch geometry. This visualization behavior was reviewed to ensure that launch conditions remained consistent across all simulation trials.